BMS COLLEGE OF ENGINEERING, BANGALORE-19

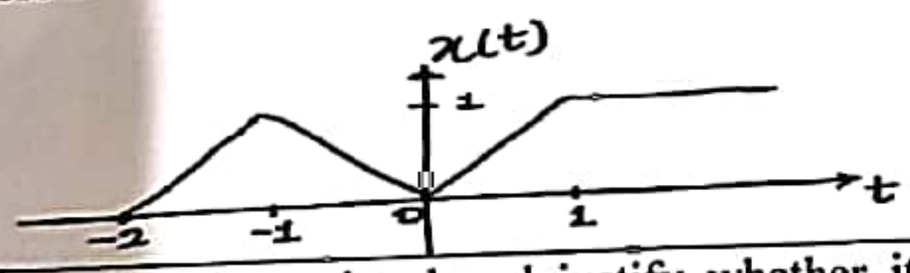
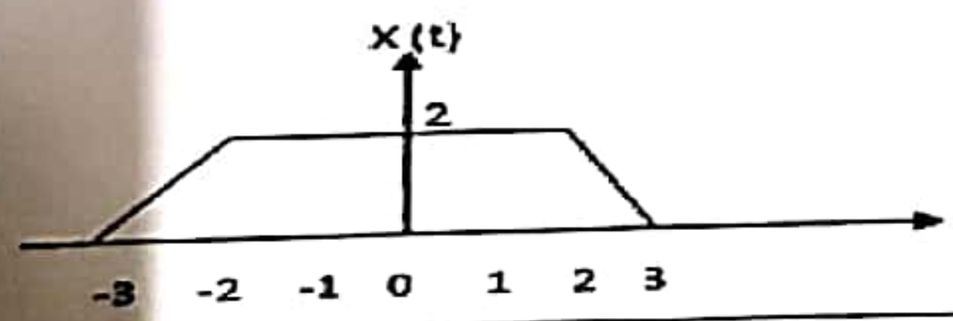
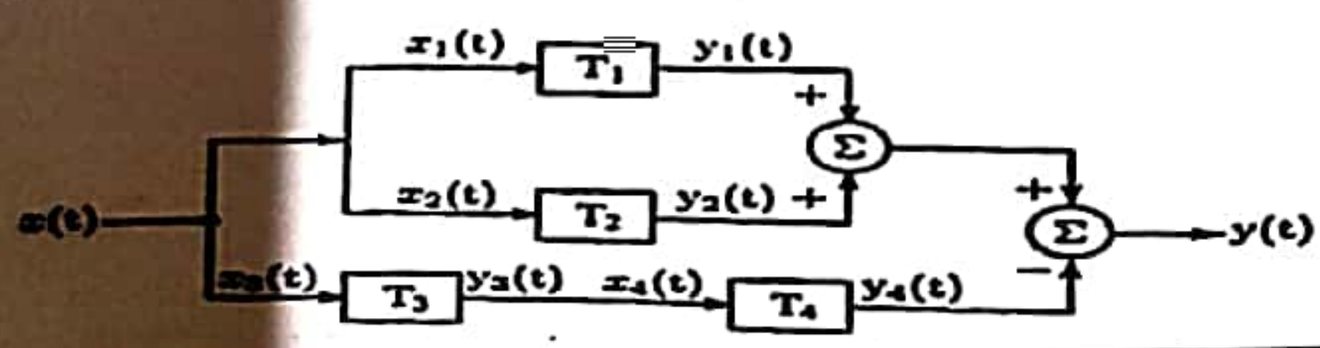
(Autonomous Institute, Affiliated to VTU)

Department of Electronics and Communication Engineering

INTERNALS -1

Course Code :	**23EC3PCSAS**	**Course Title : Signals and Systems**	
Semester :	**III**	**Maximum Marks: 40**	**Date:**
Faculty Handling the Course:		**Dr. M. A. Sunil, Prof. Shrisha M R, Dr. Feroz Morab, Prof. Suprith K S**	

PART-II

No	Question	Marks	CO	PO
1	Determine whether the following continuous time signal is periodic (or) not. If periodic, find the fundamental period i) $x(t) = [cos(2\pi t)]^2$ ii) $x(n) = 5\ sin\left(\frac{7\pi}{12}n\right) + 8\ cos\left(\frac{14\pi}{8}n\right)$ iii) $x(n) = cos\ (2n)$ iv) $x(t) = e^{j\pi t}$	10	1	1
2	Sketch the even and odd parts of the signal: a) Discrete signal $x(n)=\alpha^n u(n)$; $0<\alpha<1$ b) Continuous time signal x(t) having amplitude of 1 as shown below 	08	1	1
3	Differentiate the given signal and justify whether it is power or energy signal. 	06	2	2
4	A system consists of several subsystems connected as shown in below fig. Find the operator T relating x(t) to y(t) for the subsystem operators given by $T1: y_1(t) = x_1(t)\ x_1(t-1)$; $T2: y_2(t) = \lvert x_2(t)\rvert$; $T3: y_3(t) = 1 + 2x_3(t)$; $T4: y_4(t) = \cos(x_4(t))$ 	10	1	1

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	ii) For the discrete time signal x(n)=(8-n).{u(n)-u(n-8)}, sketch the following a) x(4-n) b) $x(\frac{n}{2}-3)$			
5	For the following system, analyze whether the system is memory or memoryless and causal, non-causal or anti causal. i) y(n)= x(2n) ii) y(t)=2t+x(t)	06	2	2

(A) (B) (C) (D) (E)

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